

10. Abandoned metal mines in Wales contribute to the pollution of some rivers and streams. The mines on Parys mountain on Anglesey have led the Afon Goch stream to become acidic and carry a range of metal ions.

- (a) The pH of the initial stretch of Afon Goch was found to be 3.24. Calculate the concentration of H^+ ions in this water.

[2]

$$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$$

- (b) The three main metals contaminating the water of Afon Goch are iron, copper and zinc. A research group wishes to find the concentrations of all three metals in a water sample from the river.

An initial reduction step converts all the iron ions present to Fe^{2+} and all the copper ions present to copper metal. The **amphoteric** zinc ions are not affected. This produces test water **A**.

Two students are asked to find the concentration of Fe^{2+} ions in test water **A**.

- (i) The reduction step produced 4.55 mg of copper metal from a 250 cm^3 sample of river water.

Suggest how this could be separated from test water **A**.

[1]

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- (ii) One student attempts to find the iron content in 25.0 cm^3 of test water **A** by gravimetric analysis. He adds excess aqueous sodium hydroxide, filters and then heats the precipitate in an oxygen-rich atmosphere.

- I. Give **two** reasons why the aqueous sodium hydroxide that is added must be in excess. [2]

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- II. The experiment produced 14.5 mg of Fe_2O_3 . Use this value to show that the concentration of Fe^{2+} ions in test water **A** is $7.27 \times 10^{-3}\text{ mol dm}^{-3}$. [2]

$$M_r(\text{Fe}_2\text{O}_3) = 159.6$$

- III. The total error when measuring the mass of Fe_2O_3 by difference was 0.2 mg .

Find the percentage error in this measurement. [1]

Percentage error = %



- (iii) The second student titrated 25.0 cm³ samples of test water **A** against a solution of acidified potassium manganate(VII) of concentration $2.10 \times 10^{-3} \text{ mol dm}^{-3}$.



The mean volume of acidified potassium manganate(VII) solution required for reaction was 18.10 cm³ and the percentage error in the titration was 0.8%.

- I. State what observation would be made at the end-point of the titration. [1]

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- II. Calculate the concentration of Fe²⁺ ions in the sample of test water **A**. [2]

Concentration = mol dm⁻³

- (iv) The two students compared their results and decided that they are in agreement.

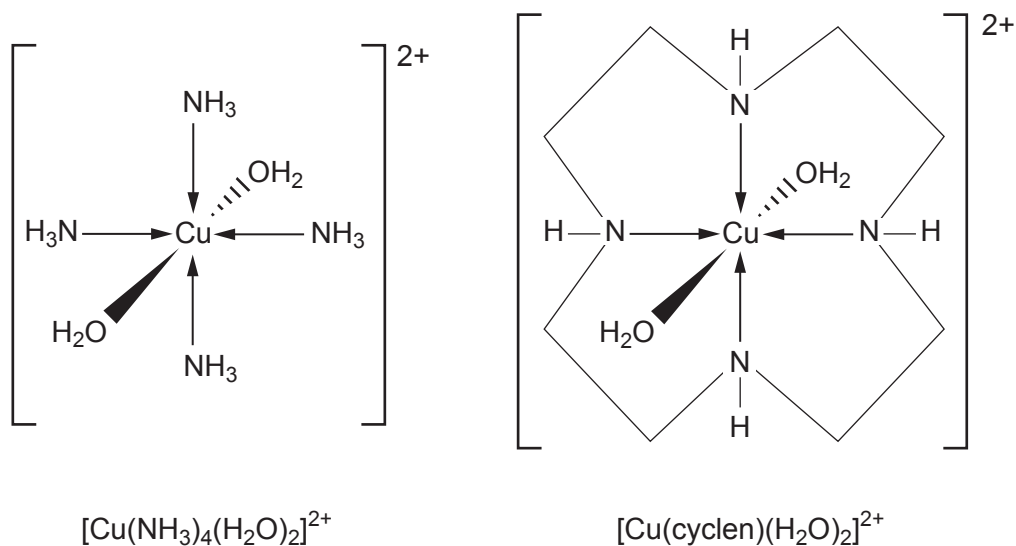
Calculate the percentage difference between the concentrations of Fe²⁺ ions found by the two students in parts (ii) and (iii) and hence show whether the students' decision is valid. [2]

Percentage difference = %

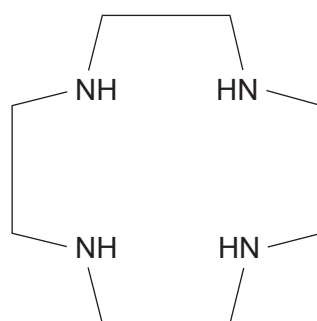
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- (c) One method of determining the concentration of copper ions present in a solution is to form a complex ion with a characteristic colour and measure the concentration of this using colorimetry. Two such copper complexes are $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ and $[\text{Cu}(\text{cyclen})(\text{H}_2\text{O})_2]^{2+}$.



Cyclen is a cyclic tetradentate ligand with four nitrogen atoms that can bond to a transition metal ion. The structure of the cyclen ligand is shown below.



cyclen

- (i) Give the colour of the $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ complex.

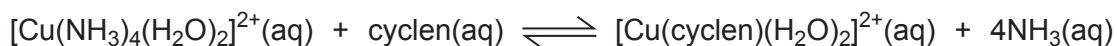
[1]

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Examiner
only

- (ii) An equilibrium exists between $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ and $[\text{Cu}(\text{cyclen})(\text{H}_2\text{O})_2]^{2+}$ in aqueous solution.



$$\Delta H^\theta = +1.2 \text{ kJ mol}^{-1}$$

Measurements show that the strengths of the Cu—N bonds in $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ and $[\text{Cu}(\text{cyclen})(\text{H}_2\text{O})_2]^{2+}$ are similar and that the Gibbs free energy change for the equilibrium is negative.

Discuss the information given above. Your answer should:

- explain why the enthalpy change for the reaction is very small
- suggest why the Gibbs free energy change for this equilibrium would be negative
- state and explain the effect of a negative Gibbs free energy change on the value of K_c for an equilibrium [6 QER]

END OF PAPER

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